

SACHA HURIOT

COMPUTER SCIENCE PHD STUDENT



- +1 314-393-7683
- sachahuriot@gmail.com
- St. Louis, MO
- sacha-huriot-tattegrain

PROFILE

I am a qualified researcher and software developer, with a M.S. in computer science alongside a diverse experience in modeling and analyzing systems. I have strong logical and creative skills. I am proficient in several programming languages, and using formal methods to verify and validate.

SKILLS

- Mathematics, logic, system modeling
- Python, Bash, SQL, Javascript, VBA
- OCaml, Coq, Lean, HLL
- Problem-solving and teamwork
- VS Code, GitHub, Slack, Emacs
- Report writing and presentation

EDUCATION

DOCTORATE IN COMPUTER SCIENCE

Washington University in St. Louis
2023 - Present

PARISIAN MASTER OF RESEARCH IN COMPUTER SCIENCE

École Normale Supérieure Paris-Saclay
2019 - 2021

BACHELOR OF FUNDAMENTAL COMPUTER SCIENCE

École Normale Supérieure Paris-Saclay
2016 - 2019

RESEARCH EXPERIENCE

RESEARCH IN SAFETY CRITICAL CONTROL

Washington University in St. Louis, United-States 2023 - Present

- Design algorithms and filters for real-time safe decision-making
- Conduct simulations for safety validation of decentralized systems

RESEARCH IN THEOREM PROVING

Vrije Universiteit Amsterdam, Netherlands 2021 - 2022

- Design and mathematical verification of Tate's algorithm
- Proving the safety of elliptic curves cryptography

RESEARCH INTERN FOR NATHALIE AUBRUN

LISN, Gif-sur-Yvette, France 2021

- Description of complexity of discrete dynamic systems
- Proving computational properties in group theory

RESEARCH INTERN FOR GLYNN WINSKEL

The Computer Lab, Cambridge, United-Kingdom 2020

- Representation of games and strategies in category theory
- Applications to concurrency and parallel computing

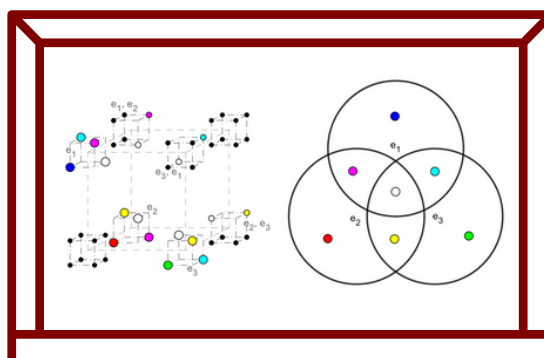
RESEARCH INTERN FOR MEHDI MHALLA

Laboratoire d'Informatique de Grenoble, France 2019

- Modeling of quantum information sharing with graphs
- Characterizing the computational power of quantum algorithms

INTERESTS

- Reading/writing science-fiction and philosophy
- Game design and development
- Digital and mathematics based art
- Science communication and education
- Creating visualizations of mathematical and programming concepts



ADDITIONAL EXPERIENCE

FORMAL METHODS

CONSULTANT 2024 - Present
Kili Technology, Remote

- Data labeling of informal mathematical reasoning
- Proof engineering in Lean 4 for machine learning

IKOS CONSULTING 2022 - 2023
Ikos Consulting, Levallois-Perret, France

- Verification of software used in critical railway reactive systems
- Specification and modeling of ground/on-board environments

**CS AND MATHEMATICS
TUTORING** 2018 - 2023
Complétude/Groupe Réussite, Paris, France

- Lessons and exercises personalized for the student's needs
- High school and undergraduate-level computer science courses

**DATABASE SOFTWARE
DEVELOPER** 2018
STM Nervi, Manosque, France

- Development of a comprehensive facture sorting software
- Front end and UI design answering the client's specifications

**FINANCIAL MODELING
SOFTWARE DEVELOPER** 2017
Coélience, Paris, France

- Development of an economic prediction tool for market analysis
- Implementing mathematical models and a user-friendly interface

PUBLICATIONS

SAFE DECENTRALIZED MULTI-AGENT CONTROL USING BLACK-BOX PREDICTORS AND CONTROL BARRIER FUNCTIONS

Hussein Sibai, Sacha Huriot
Submitted to ICRA 2025, arXiv:2409.18862

SYMMETRY-BASED ABSTRACTION ALGORITHM FOR ACCELERATING SYMBOLIC CONTROL SYNTHESIS

Hussein Sibai, Sacha Huriot, Tyler Martin, Murat Arcak
NFM 2024, arXiv:2403.11816

APERIODICITY AND DOMINO PROBLEMS ON FINITELY PRESENTED GROUPS

Nathalie Aubrun, Nicolas Bitar, Sacha Huriot
MFCS 2022, doi.org/10.1017/etds.2023.44

CONCURRENT STRATEGIES ON GAMES WITH ALGEBRAS

Sacha Huriot, Glynn Winskel
preprint 2024, arXiv:2402.04590

CONTEXTUALITY AND EXPRESSIVITY OF NON-LOCALITY

Sacha Huriot, Mehdi Mhalla
EPTCS QPL 2020, arXiv:1912.11412

